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# 廖 振宇

## 工作经历

2021–至今 **副研究员**, 华中科技大学, 电子信息与通信学院

2020–2021 **博士后研究员**, 加州大学伯克利分校, 统计系和ICSI，合作导师：Michael Mahoney教授

## 教育经历

2016–2019 **博士**, 巴黎萨克雷大学, 数学与计算机，CentraleSupélec；导师：Romain Couillet教授

2014–2016 **硕士**, 巴黎萨克雷大学, 信号与图像处理，导师：Romain Couillet教授

2010–2014 **本科**, 华中科技大学, 光电信息工程

## 荣誉与奖励

2026 NeurIPS 2026 Top AC（优秀领域主席）

2026 Springer Nature 杰出编辑——*2026 年度作者服务奖*（作为Statistics and Computing编委）

2025 IEEE 高级会员

2025 MATRIX-Simons Travel Grant，MATRIX，Creswick，澳大利亚

2025 加拿大 CRM-Simons 访问教授，蒙特利尔大学数学研究中心 “Mathematical Foundations of Data Science” 专题研究项目（签证原因未成行）

2024 法国 ANR-LabEx-CIMI 访问教授，图卢兹第三大学

2023 湖北省人才计划

2021 武汉市人才计划

2021 华中科技大学东湖青年学者

2019 巴黎萨克雷大学ED STIC 优秀博士论文

## 学术专著

**CUP 2022** Romain Couillet and **Zhenyu Liao** (2022). Random Matrix Methods for Machine Learning. Cambridge University Press. DOI: 10.1017/9781009128490.

## 代表论文

### 会议

**NeurIPS 2026** Chengmei Niu, Sachin Garg, Michal Derezinski, and **Zhenyu Liao** (2026). Debiasing Random Oblique Projections for Subsampled OLS and Fast CUR in High Dimensions. *Advances in Neural Information Processing Systems (**NeurIPS**)*.

**ECML 2026** Yanlei Liu and **Zhenyu Liao** (Sept. 2027). Dual-Attention Convolution Experts for Sparse Tensor Completion. *Machine Learning and Knowledge Discovery in Databases. Research Track (**ECML PKDD**)*. Springer Nature Switzerland, pp. 450–467. URL: https://link.springer.com/chapter/10.1007/978-3-032-37670-1_26.

**ICML 2026** Mohamed Chiheb Yaakoubi, Cosme Louart, Malik Tiomoko, and **Zhenyu Liao** (July 2026). Characterization of Gaussian Universality Breakdown in High-Dimensional Empirical Risk Minimization. *Proceedings of the 43rd International Conference on Machine Learning (**ICML**)*. Vol. 306. PMLR, pp. 143332–143358. URL: https://proceedings.mlr.press/v306/yaakoubi26a.html.

**ICML 2026** Jiayu Bai, Danchen Yu, **Zhenyu Liao**, Tianqi Hou, Feng Zhou, Robert C. Qiu, and Zenan Ling (July 2026). Diving into Kronecker Adapters: Component Design Matters. *Proceedings of the 43rd International Conference on Machine Learning (**ICML**)*. Vol. 306. PMLR, pp. 5479–5501. URL: https://proceedings.mlr.press/v306/bai26l.html.

**EUSIPCO 2026** Yue Xu, Shuo Zhang, Gui Zhou, and **Zhenyu Liao** (2026). Optimal Large-scale Finite-horizon Optimization via Random Matrix Theory. *2026 34th European Signal Processing Conference (**EUSIPCO**)*.

**ICASSP 2026** Yue Xu and **Zhenyu Liao** (2026b). An Improved Convergence Analysis of Gossip Methods for Large Random Graphs. *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**)*, pp. 22562–22566. DOI: 10.1109/ICASSP55912.2026.11461477.

**ICASSP 2026** Yessin Moahker, Malik Tiomoko, Cosme Louart, and **Zhenyu Liao** (2026). A Random Matrix Perspective of Echo State Networks: From Precise Bias-Variance Characterization to Optimal Regularization. *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**)*, pp. 20716–20720. DOI: 10.1109/ICASSP55912.2026.11464627.

**SIGMOD 2026** Jiuqi Wei, **Zhenyu Liao**, Ruoyu Han, Quanqing Xu, Chuanhui Yang, and Themis Palpanas (May 2026). TaCo: Data-adaptive and Query-aware Subspace Collision for High-dimensional Approximate Nearest Neighbor Search. *Proc. ACM Manag. Data (**SIGMOD**)* **4**(3). DOI: 10.1145/3802118. URL: https://doi.org/10.1145/3802118.

**ICML 2025** Chengmei Niu, **Zhenyu Liao**, Zenan Ling, and Michael W. Mahoney (July 2025). Fundamental Bias in Inverting Random Sampling Matrices with Application to Sub-sampled Newton. *Proceedings of the 42nd International Conference on Machine Learning (**ICML**)*. Vol. 267. PMLR, pp. 46649–46692. URL: https://proceedings.mlr.press/v267/niu25c.html.

**ICLR 2025** Xiaoyi Mai and **Zhenyu Liao** (2025). The Breakdown of Gaussian Universality in Classification of High-dimensional Linear Factor Mixtures. *International Conference on Learning Representations (**ICLR**)*. URL: https://openreview.net/forum?id=UrKbn51HjA.

**SIGMOD 2025** Jiuqi Wei, Xiaodong Lee, **Zhenyu Liao**, Themis Palpanas, and Botao Peng (Feb. 2025). Subspace Collision: An Efficient and Accurate Framework for High-dimensional Approximate Nearest Neighbor Search. *Proc. ACM Manag. Data (**SIGMOD**)* **3**(1). DOI: 10.1145/3709729. URL: https://doi.org/10.1145/3709729.

**EUSIPCO 2024** Wei Yang, Zhengyu Wang, Xiaoyi Mai, Zenan Ling, Robert Caiming Qiu, and **Zhenyu Liao** (Aug. 2024). Inconsistency of ESPRIT DoA Estimation for Large Arrays and a Correction via RMT. *2024 32nd European Signal Processing Conference (**EUSIPCO**)*, pp. 2722–2726. DOI: 10.23919/EUSIPCO63174.2024.10715081.

**ICML 2024** Zenan Ling, Longbo Li, Zhanbo Feng, Yixuan Zhang, Feng Zhou, Robert C. Qiu, and **Zhenyu Liao** (July 2024). Deep Equilibrium Models Are Almost Equivalent to Not-so-deep Explicit Models for High-dimensional Gaussian Mixtures. *Proceedings of the 41st International Conference on Machine Learning (**ICML**)*. Vol. 235. PMLR, pp. 30585–30609. URL: https://proceedings.mlr.press/v235/ling24a.html.

**NeurIPS 2022** Lingyu Gu, Yongqi Du, Yuan Zhang, Di Xie, Shiliang Pu, Robert Qiu, and **Zhenyu Liao** (2022). "Lossless" Compression of Deep Neural Networks: A High-dimensional Neural Tangent Kernel Approach. *Advances in Neural Information Processing Systems (**NeurIPS**)*. Vol. 35. Curran Associates, Inc., pp. 3774–3787. URL: https://proceedings.neurips.cc/paper_files/paper/2022/hash/185087ea328b4f03ea8fd0c8aa96f747-Abstract-Conference.html.

**ICLR 2022** Hafiz Tiomoko Ali, **Zhenyu Liao**, and Romain Couillet (2022). Random matrices in service of ML footprint: ternary random features with no performance loss. *International Conference on Learning Representations (**ICLR**)*. URL: https://openreview.net/forum?id=qwULHx9zld.

**NeurIPS 2021** **Zhenyu Liao** and Michael W Mahoney (2021). Hessian Eigenspectra of More Realistic Nonlinear Models. *Advances in Neural Information Processing Systems (**NeurIPS**)*. Vol. 34. Curran Associates, Inc., pp. 20104–20117. URL: https://papers.nips.cc/paper/2021/hash/a7d8ae4569120b5bec12e7b6e9648b86-Abstract.html.

**COLT 2021** Michal Derezinski, **Zhenyu Liao**, Edgar Dobriban, and Michael Mahoney (Aug. 2021). Sparse sketches with small inversion bias. *Proceedings of Thirty Fourth Conference on Learning Theory (**COLT**)*. Vol. 134. PMLR, pp. 1467–1510. URL: https://proceedings.mlr.press/v134/derezinski21a.html.

**ICLR 2021** **Zhenyu Liao**, Romain Couillet, and Michael W. Mahoney (2021a). Sparse Quantized Spectral Clustering. *International Conference on Learning Representations (**ICLR**)*. URL: https://openreview.net/forum?id=pBqLS-7KYAF.

**AISTATS 2021** Fanghui Liu, **Zhenyu Liao**, and Johan Suykens (Apr. 2021). Kernel Regression in High Dimension: Refined Analysis beyond Double Descent. *Proceedings of The 24th International Conference on Artificial Intelligence and Statistics (**AISTATS**)*. Vol. 130. PMLR, pp. 649–657. URL: http://proceedings.mlr.press/v130/liu21b.html.

**NeurIPS 2020** **Zhenyu Liao**, Romain Couillet, and Michael W. Mahoney (2020). A Random Matrix Analysis of Random Fourier Features: Beyond the Gaussian Kernel, A Precise Phase Transition, and the Corresponding Double Descent. *Advances in Neural Information Processing Systems (**NeurIPS**)*. Vol. 33. Curran Associates, Inc., pp. 13939–13950. URL: https://papers.nips.cc/paper/2020/hash/a03fa30821986dff10fc66647c84c9c3-Abstract.html.

**NeurIPS 2020** Michal Derezinski, Feynman T Liang, **Zhenyu Liao**, and Michael W. Mahoney (2020). Precise expressions for random projections: Low-rank approximation and randomized Newton. *Advances in Neural Information*

*Processing Systems (**NeurIPS**)*. Vol. 33. Curran Associates, Inc., pp. 18272–18283. URL: https://papers.nips.cc/paper/2020/hash/d40d35b3063c11244fbf38e9b55074be-Abstract.html.

**ICASSP 2019** Xiaoyi Mai, **Zhenyu Liao**, and Romain Couillet (May 2019). A Large Scale Analysis of Logistic Regression: Asymptotic Performance and New Insights. *IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**)*. IEEE, pp. 3357–3361. DOI: 10.1109/ICASSP.2019.8683376.

**EUSIPCO 2018** Romain Couillet, **Zhenyu Liao**, and Xiaoyi Mai (Sept. 2018). Classification Asymptotics in the Random Matrix Regime. *The 26th European Signal Processing Conference (**EUSIPCO**)*. IEEE, pp. 1875–1879. DOI: 10.23919/EUSIPCO.2018.8553034.

**ICML 2018** **Zhenyu Liao** and Romain Couillet (July 2018a). The Dynamics of Learning: A Random Matrix Approach. *Proceedings of the 35th International Conference on Machine Learning (**ICML**)*. Vol. 80. PMLR, pp. 3072–3081. URL: http://proceedings.mlr.press/v80/liao18b.html.

**ICML 2018** **Zhenyu Liao** and Romain Couillet (July 2018b). On the Spectrum of Random Features Maps of High Dimensional Data. *Proceedings of the 35th International Conference on Machine Learning (**ICML**)*. Vol. 80. PMLR, pp. 3063–3071. URL: http://proceedings.mlr.press/v80/liao18a.html.

**ICASSP 2017** **Zhenyu Liao** and Romain Couillet (Mar. 2017a). Random Matrices Meet Machine Learning: A Large Dimensional Analysis of LS-SVM. *IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**)*. IEEE, pp. 2397–2401. DOI: 10.1109/ICASSP.2017.7952586.

### 期刊

**SPM 2026** **Zhenyu Liao** and Michael W. Mahoney (2026). Random Matrix Theory for Deep Learning: Beyond Eigenvalues of Linear Models. *IEEE Signal Processing Magazine* **43**(2), 93–106. DOI: 10.1109/MSP.2025.3618012.

**TSP 2026** Zhengyu Wang, Wei Yang, Xiaoyi Mai, Zenan Ling, **Zhenyu Liao**, and Robert C. Qiu (2026). A Large-Dimensional Analysis of ESPRIT DoA Estimation: Inconsistency and a Correction via RMT. *IEEE Transactions on Signal Processing* **74**, 2290–2303. DOI: 10.1109/TSP.2026.3683673.

**TPAMI 2026** Junchao Lin, Zenan Ling, Zhanbo Feng, Jingwen Xu, Minxuan Liao, Feng Zhou, Tianqi Hou, **Zhenyu Liao**, and Robert C. Qiu (2026). IGNN-Solver: A Graph Neural Solver for Implicit Graph Neural Networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 1–18. DOI: 10.1109/TPAMI.2026.3729092.

**TKDE 2024** Zhanbo Feng, Yuanjie Wang, Jie Li, Fan Yang, Jiong Lou, Tiebin Mi, Robert. C. Qiu, and **Zhenyu Liao** (2025). Robust and Communication-Efficient Federated Domain Adaptation via Random Features. *IEEE Transactions on Knowledge and Data Engineering* **37**(3), 1411–1424. DOI: 10.1109/TKDE.2024.3510296.

**MCRF 2022** Yacine Chitour, **Zhenyu Liao**, and Romain Couillet (2023). A geometric approach of gradient descent algorithms in linear neural networks. *Mathematical Control and Related Fields* **13**(3), 918–945. DOI: 10.3934/mcrf.2022021.

**JSTAT 2021** **Zhenyu Liao**, Romain Couillet, and Michael W. Mahoney (Dec. 2021b). A random matrix analysis of random Fourier features: beyond the Gaussian kernel, a precise phase transition, and the corresponding double descent. *Journal of Statistical Mechanics: Theory and Experiment* **2021**(12), 124006. DOI: 10.1088/1742-5468/ac3a77. URL: https://doi.org/10.1088/1742-5468/ac3a77.

**TSP 2019** **Zhenyu Liao** and Romain Couillet (Feb. 2019b). A Large Dimensional Analysis of Least Squares Support Vector Machines. *IEEE Transactions on Signal Processing* **67**(4), 1065–1074. DOI: 10.1109/TSP.2018.2889954.

**AAP 2018** Cosme Louart, **Zhenyu Liao**, and Romain Couillet (Apr. 2018). A Random Matrix Approach to Neural Networks. *The Annals of Applied Probability* **28**(2), 1190–1248. DOI: 10.1214/17-AAP1328. URL: https://doi.org/10.1214/17-AAP1328.

## 科研项目

### 在研

2026–2029 国家自然科学基金面上项目：基于随机矩阵方法的 *Transformer* 大模型压缩：从缩放定律极限理论到动态自适应模型压缩方法（NSFC-12571561），44 万元，**主持**；Co-PI：Cosme Louart，香港中文大学（深圳）

2025–2030 国家重点研发计划青年科学家项目：揭示大模型机理的机器学习基础理论研究（2025YFA1018600），100/300 万元，项目骨干；PI：束俊，西安交通大学

2026–2028 华中科技大学“基础研究支持计划”数学专项项目：基于随机矩阵方法的大规模机器学习理论和方法（2025BRSXB0004），38 万元，**主持**

2024–2026 华为技术有限公司校企合作项目：*Algorithm and Theory of High-dimensional Dynamical Systems*（TC20240702035），80 万元，**主持**；Co-PI：凌泽南，华中科技大学

### 已结题

2024–2026 广东省人工智能数理基础重点实验室开放基金：基于随机矩阵方法的 *Transformer* 模型泛化理论研究（OFA00003），10 万元，**主持**；Co-PI：Jeff Yao，香港中文大学（深圳）

2023–2025 国家自然科学基金青年科学基金项目：基于随机矩阵方法的神经网络模型剪枝基础理论研究（NSFC-62206101），30 万元，**主持**

2023–2025 华为技术有限公司校企合作项目：随机矩阵理论驱动的通信理论和算法研究（TC20231122043），59 万元，**主持**

2022–2025 国家自然科学基金“面向未来通信的数学基础（信息论）”专项项目：智能反射面辅助的新型无线通信数学理论与数学技术（NSFC-12141107），30/300 万元，项目骨干；PI：邱才明、段金桥，华中科技大学

2021–2022 中国计算机学会 CCF-海康威视斑头雁基金项目：基于随机矩阵和信息瓶颈理论的神经网络表达和压缩的研究（20210008），28 万元，**主持**；Co-PI：万凯，华中科技大学

2018–2021 NSF Research Grant，*Combining Stochastics and Numerics for Improved Scalable Matrix Computations*（NSF-1815054），50 万美元，项目骨干；PI：Michael W. Mahoney

2018–2021 法国高等教育、研究与创新部：GSTATS-IDEX DataScience Chair，30 万欧元，项目骨干；PI：Romain Couillet

2014–2017 法国国家科研署项目：*Random Matrix Theory for Large Dimensional Graphs*（ANR-14-CE28-0006），30 万欧元，项目骨干；PI：Romain Couillet

## 教学改革项目

2024-2025 华中科技大学国家卓越工程师学院 2025 年“卓越工程师自主培养新范式”项目：产教融合卓越工程师国际化视野培养，3 万元，**主持**（已结题）

## 科研服务

**学会兼职** 中国现场统计研究会随机矩阵理论与应用分会副秘书长；中国现场统计研究会大数据统计分会理事；中国商业统计学会人工智能分会理事；中国通信学会短距无线通信委员会秘书

**国际会议学术职务** ICML 2025–2026、NeurIPS 2025–2026、ICLR 2025–2026、AISTATS 2026 和 IJCNN 2025–2026 **领域主席（AC）**；AAAI 2027**高级程序委员会委员（SPC）**

**国际期刊编委** Springer Statistics and Computing编委（2025 年至今）；SCIENCE CHINA Mathematics“随机矩阵理论及其应用”专题客座编辑

**学术活动组织**

1. 与姚建峰、杨帆、胡江、姜丹丹和郑术蓉共同组织2027 年随机矩阵理论与应用年会，2027 年 1 月 14 日至 17 日，海南三亚
2. 与鲍志刚、Benoît Collins、姚东、Jeff Yao和郑术蓉共同组织天元国际数学交流中心随机矩阵前沿理论与应用（TMRC-FRMTA）研讨会，2026 年 8 月 16 日至 22 日，云南昆明
3. 2026 IEEE/CIC International Conference on Communications in China (ICCC)**赞助主席**，2026 年 8 月 7 日至 9 日
4. 与束俊共同组织2026 中国机器学习与科学应用大会（CSML 2026）“现代机器学习理论与方法”专题，2026 年 8 月 7 日至 9 日，上海
5. 与刘方辉、吴磊共同组织CSIAM 第七届大数据与人工智能科学大会（CSIAM-BDAI 2026）TM10“大模型的理论指导实践”专题，2026 年 7 月 17 日至 19 日，福建厦门
6. 与Rishabh Dudeja、马俊杰和Arian Maleki共同组织 ISIT 2026 Universality and Dynamics in High-Dimensional Learning and Inference Workshop，2026 年 7 月 3 日，广东广州
7. 与Dirk Slock共同组织 ICASSP 2026 特别分会 *Large-Dimensional Structures and Methods for Signal Processing, Communication, and Machine Learning*，2026 年 5 月，西班牙巴塞罗那
8. 与杨朋昆和方聪共同组织2025 年大模型数理统计基础研讨会，2025 年 7 月，吉林长白山
9. 联合组织中国现场统计研究会随机矩阵理论与应用分会成立大会，2025 年 1 月，吉林长春
10. 第十届 EAI 物联网即服务国际会议（EAI IoTaaS 2024）**大会共同主席**，湖北武汉
11. 与 Mihai Nica、Courtney Paquette、Elliot Paquette、Andrew Saxe 和 René Vidal 共同组织首届 ICML 2023 High-dimensional Learning Dynamics (HiLD) Workshop
12. 与 Yacine Chitour 共同组织华中科技大学–巴黎萨克雷大学 2022 联合研讨会“Math for Data Science”；超过 4 万名在线参与者

**学位论文评审** 受邀担任澳大利亚昆士兰大学博士学位论文外审专家（2026 年）；受邀担任清华大学博士学位论文外审专家（2021 年）

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| **科研项目评审专家** | 欧洲研究理事会（ERC），加拿大自然科学与工程研究委员会（NSERC），以色列科学基金会（ISF），中国国家自然科学基金委员会（NSFC） |
| **国际会议审稿人** | ICLR，ICML，NeurIPS，SODA，AISTATS，AAAI，ECAI，IEEE-ICASSP，IEEE-CAMSAP |
| **数学与数据科学期刊审稿人** | SIAM Review (SIREV)，Foundations of Computational Mathematics (FoCM)，Physical Review X (PRX)，IEEE Transactions on Information Theory (TIT)，Journal of Mathematical Physics，Springer Statistics and Computing (STCO)，SIAM Journal on Scientific Computing (SISC)，Random Matrices: Theory and Applications (RMTA)，Latin American Journal of Probability and Mathematical Statistics (ALEA)，Annals of Applied Probability (AAP)，Electronic Communications in Probability (ECP)，Electronic Journal of Statistics (EJS)，Journal of Applied Mathematics and Computing (JAMC)，AIMS Mathematics |
| **机器学习和 AI 期刊审稿人** | Journal of Machine Learning Research (JMLR)，IEEE Transactions on Pattern Analysis and Machine Intelligence (IEEE-TPAMI)，IEEE Transactions on Dependable and Secure Computing (TDSC)，IEEE Signal Processing Letters (SPL)，IEEE Transactions on Signal Processing (IEEE-TSP)，IEEE Transactions on Multimedia (IEEE-TMM)，Engineering Applications of Artificial Intelligence (EAAI)，Frontiers of Computer Science，Neural Networks (NEUNET)，IEEE Transactions on Neural Networks and Learning Systems (IEEE-TNNLS)，IEEE Transactions on Industrial Informatics (IEEE-TII)，Transactions on Machine Learning Research (TMLR)，Pattern Recognition (PR)，Neural Processing Letters (NPL) |

## 推荐人


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|---|---|
| **邱才明** | 华中科技大学电子信息与通信学院院长，IEEE Fellow。Email: caiming@hust.edu.cn |
| **Michael W. Mahoney** | Professor in the Department of Statistics, UC Berkeley, and Director of the UC Berkeley FODA (Foundations of Data Analysis) Institute, Berkeley, CA, USA. Email: mmahoney@stat.berkeley.edu |
| **Romain Couillet** | Full Professor at University Grenoble-Alps, France. Holder of the UGA MIAI LargeDATA Chair, University-Grenoble-Alps, France. Email: romain.couillet@gipsa-lab.grenoble-inp.fr |